

(a) Framework illustrating the practical application of the data

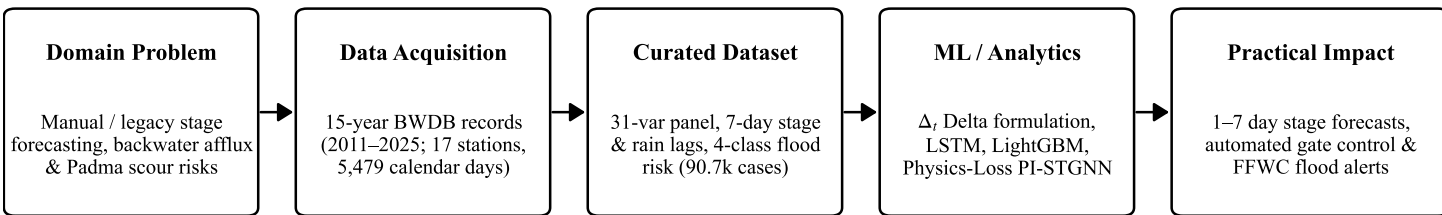


Figure 1(a). Dataset-specific practical application framework illustrating end-to-end impact from hydrometric collection to automated river corridor warning and bridge operations.

(b) Technical pipeline of the ML method using the dataset

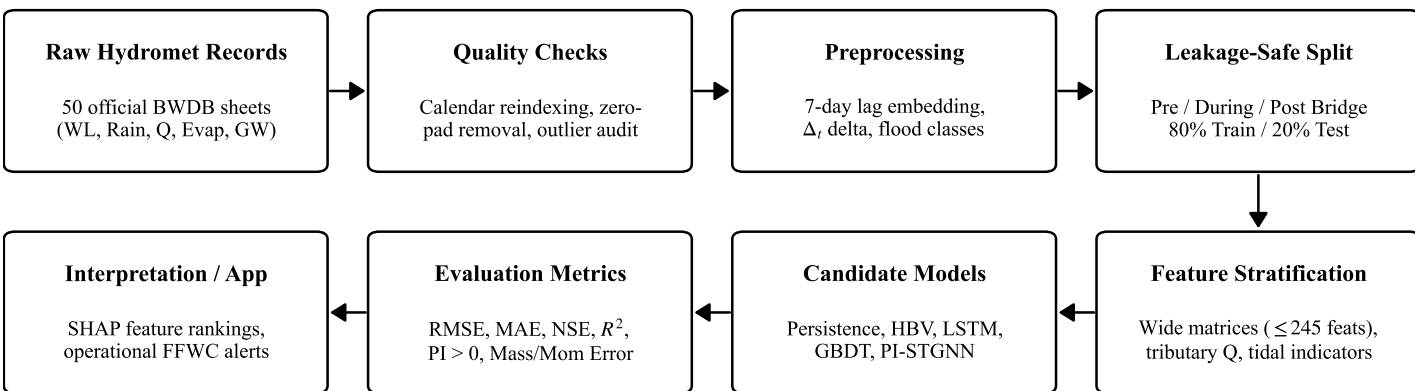


Figure 1(b). Exact technical pipeline showing leakage-safe partitioning, multivariate lag matrix generation, physics-guided architectures, and persistence-benchmarked evaluation.